

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                      | Marks available |     |     |       |       |      |
|----------|-----|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                      | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7        | (a) |      |  | award (1) for either of following <ul style="list-style-type: none"> <li>prevents acid (-spray) escaping</li> <li><u>only</u> allows carbon dioxide to escape</li> </ul>                                                                                                                                                                                             |                 |     | 1   | 1     |       | 1    |
|          | (b) |      |  | take the mass (after 10 s) away from 107.75 / the original mass                                                                                                                                                                                                                                                                                                      |                 |     | 1   | 1     |       | 1    |
|          | (c) |      |  | 0.0082 / $8.2 \times 10^{-3}$ (3) <b>must</b> be to 2 sig figs<br><br>if incorrect award (1) each for either of following<br><br>appropriate <b>change</b> in mass e.g. 0.44 – 0.30 / 0.14<br><br>corresponding <b>change</b> in time e.g. 30 – 13 / 17<br><br>ecf possible from error in graph reading but must have calculated a change in mass and change in time |                 | 3   |     | 3     | 3     |      |
|          | (d) | (i)  |  | award (2) for all points plotted correctly – tolerance $\pm \frac{1}{2}$ square<br>award (1) for any four points plotted correctly<br><br>award (1) for (smooth) curve drawn through points<br><br>ecf possible from incorrectly plotted points                                                                                                                      |                 | 3   |     | 3     | 3     |      |
|          |     | (ii) |  | acid has lower concentration (1)<br><br>so fewer particles present (in the same volume) (1)<br><br>lower chance of successful collisions / lower frequency of successful collisions / fewer successful collisions (per second) (1)                                                                                                                                   | 2               |     | 1   | 3     |       |      |